# Overview

This class is about **probabilistic modeling**, which means formalizing our assumptions and knowledge about how the stuff we observe (i.e., data) relates to the things we want to infer using the mathematical language of probability.

# Two-class problem

Let's consider perhaps the simplest modeling problem there is: the two-class problem.

$$y \in \{0,1\} \text{ disease} \qquad x \in \{0,1\} \text{ diagnostic}$$

Say we are in a hospital setting and interested in diagnosing whether patients have a particular disease (y=1) or not (y=0). A cheap diagnostic test which outputs positive (x=1) or negative (x=0) is always available, but not necessarily always accurate about the true diagnosis.

Viewing each patient i as a random iid sample from a population-of-interest, say our data are:

$$(x_i, y_i) \overset{iid}{\sim} \underbrace{\Pr(x, y)}_{\text{population frequency}}, \qquad i = 1 \ldots n$$

Meaning that for n patients we run the cheap diagnostic x but also run a more expensive procedure to actually measure the true diagnosis y. We will not always be able to observe y.

We distinguish the population frequency from the sample frequency, defining the latter as:

$$\underbrace{p(x, y)}_{\text{sample frequency}} \overset{def}{=} \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}(x_i = x)\, \mathbb{1}(y_i = y)$$

For a new unseen patient, we now want to use a model to predict y from x.

Consider the following two "models":

$$p(x \mid y=0) \qquad \text{vs.} \qquad p(x \mid y=1)$$

We observe x, and want to predict y. So we can compare the "likelihood" of y=1 vs y=0. One way to do that is via the likelihood ratio (LR), also sometimes called the "Bayes factor":

$$\frac{p(x \mid y=1)}{p(x \mid y=0)} \overset{e.g.}{=} 3$$

An LR=3 would mean that "the data x are 3 times more likely under y=1" or in other words, the "the data x are 3-to-1 in favor of the hypothesis y=1".

So should we conclude that y=1? Not so fast. What if the disease is extremely rare? The likelihood ratio above does not account for the "base rate" of the disease in the population:

$$pr(y=1) \overset{e.g.}{=} 0.01$$

We can treat the base rate in our sample as a kind of "**prior distribution**":

$$p(y=1) \overset{def}{=} \frac{1}{n}\sum_{i=1}^{n} y_i$$

That is, prior to seeing a patient's diagnostic test x, our best guess for their diagnosis is just the estimated base rate of the disease.

Once we observe a patient's diagnostic x, the correct probability to query is then the **posterior distribution**, which is given by Bayes' rule:

$$\underbrace{p(y=1 \mid x)}_{\text{posterior}} = \frac{\overbrace{p(x \mid y=1)}^{\text{likelihood}}\ \overbrace{p(y=1)}^{\text{prior}}}{\underbrace{p(x)}_{\text{evidence}}}$$

The denominator on the RHS is the "model evidence" or the "marginal likelihood":

$$p(x) = \sum_y p(x, y) = \sum_x p(x \mid y)\, p(y)$$

In this case, since both x and y takes only two values, the evidence is easily calculable...

$$p(x=1) = p(x=1, y=1) + p(x=1, y=0)$$

$$p(x=0) = 1 - p(x=1)$$

...so the posterior distribution can be easily calculated as well. This is not always true.

The posterior says "given a diagnostic test x, what is the probability of y=1" while the likelihood says "given y=1, what is the probability of x?" The posterior takes the base rate (or prior distribution) of y into account.

One way to view the posterior in the case of a binary y is via the **posterior odds**:

$$\underbrace{\frac{p(y=1 \mid x)}{p(y=0 \mid x)}}_{\text{posterior odds}} = \underbrace{\frac{p(x \mid y=1)}{p(x \mid y=0)}}_{\text{likelihood ratio}} \times \underbrace{\frac{p(y=1)}{p(y=0)}}_{\text{prior odds}}$$

As an example, consider a rare disease, where the diagnostic has a true positive rate (TPR) of 1.0 and a false positive rate (FPR) of 0.1:

e.g. $p(x=1 \mid y=1) = 1.0$

$p(x=1 \mid y=0) = 0.1$

$p(y=1) = 0.001$

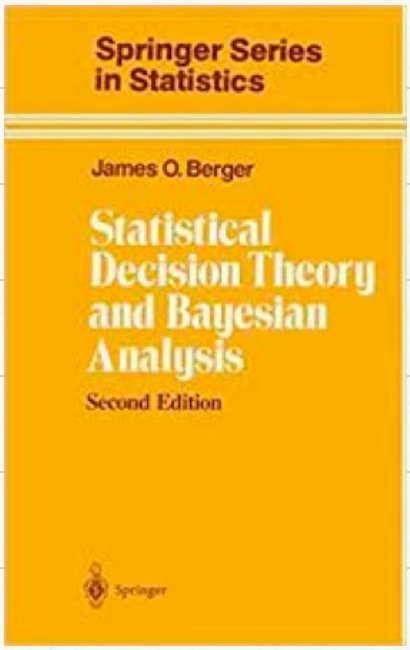
$$\longrightarrow \quad \frac{p(y=1 \mid x)}{p(y=0 \mid x)} = \frac{1.0}{0.1} \times \frac{0.001}{0.999} \approx 0.01$$

In this case, while the likelihood ratio is 10-to-1 in favor of the hypothesis y=1, the posterior odds are about 100-to-1 against! Base rates matter.

Note this is not a mere issue of **statistical significance**. Consider the classic "lady drinking tea" example from Berger (1985).

1. A lady, who adds milk to her tea, claims to be able to tell whether the tea or the milk was poured into the cup first. In all of ten trials conducted to test this, she correctly determines which was poured first.
2. A music expert claims to be able to distinguish a page of Haydn score from a page of Mozart score. In ten trials conducted to test this, he makes a correct determination each time.
3. A drunken friend says he can predict the outcome of a flip of a fair coin. In ten trials conducted to test this, he is correct each time.

In all three situations, the unknown quantity $\theta$ is the probability of the person answering correctly. A classical significance test of the various claims would consider the null hypothesis ($H_0$) that $\theta = 0.5$ (i.e., the person is guessing). In all three situations this hypothesis would be rejected with a (one-tailed) significance level of $2^{-10}$. Thus the above experiments give strong evidence that the various claims are valid.




It's not just base rates that matter, it's also **prior beliefs** (more on this later).

# Decision theory

So base rates or priors matter for predicting y from x. What else matters in this setting? Say the posterior odds are 100-to-1 in favor of y=1. Should we treat the patient? Again, not so fast.

We have to consider the cost (or loss) associated with each decision. Define the **decision rule** and the **loss function**:

| decision rule | loss function |
|---|---|
| $\hat{y}(x) \in \{0,1\}$ | $\ell(y, \hat{y}(x)) \in \mathbb{R}$ |

For binary classification problem the loss function defines four numbers corresponding to the losses for true negatives (TN), false negatives (FN), false positives (FP), and true positives (TP):

| | $\hat{y}(x)=0$ | $\hat{y}(x)=1$ |
|---|---|---|
| $y=0$ | $\ell(0,0)$ TN | $\ell(0,1)$ FP |
| $y=1$ | $\ell(1,0)$ FN | $\ell(1,1)$ TP |

$$\ell_{TN}, \ell_{TP} \geq \ell_{FN}, \ell_{FP}$$

The loss is something we hope to minimize. Define the **integrated risk**:

$$r(\hat{y}(\cdot)) = \mathbb{E}_{x,y}\left[\ell(y, \hat{y}(x))\right]$$

$$\equiv \mathbb{E}_{(x,y)\sim p_r(x,y)}\left[\ell(y, \hat{y}(x))\right]$$

This is the loss averaged over the population of (x,y) pairs. The optimal decision rule is then:

$$\hat{y}^*(\cdot) = \underset{\hat{y}(\cdot)}{\operatorname{argmin}}\ r(\hat{y}(\cdot))$$

$$= \underset{\hat{y}(\cdot)}{\operatorname{argmin}}\ \mathbb{E}_{x}\ \mathbb{E}_{y|x=x}\left[\ell(y, \hat{y}(x))\right]$$

The optimal rule thus "always takes the right action" for any input x:

$$\hat{y}^*(x) = \underset{a}{\operatorname{argmin}}\ \mathbb{E}_{y|x=x}\left[\ell(y, a)\right]$$

In other words, we only need to know Pr(y | x) in order to obtain the optimal decision rule.
As a result, the optimal decision rule is often called the "**Bayes decision rule**" and its risk is called the "**Bayes risk**"

**Bayes decision rule:** $\hat{y}^*(\cdot)$ **Bayes risk:** $r(\hat{y}^*(\cdot))$

This is because the term "Bayesian inference" usually means "posterior inference": computing a posterior distribution, given assumptions (or knowledge) of a likelihood and a prior. However, here note that the integrated risk is a **frequentist** criterion, as it marginalizes over the **population distribution** Pr(x, y). The optimality of the Bayes decision rule is often cited as frequentist justification for Bayesian methods (though that is not very good justification, in my opinion).

## Modeling: Discriminative vs generative

We almost never know the population posterior Pr(y | x). Moreover obtaining a good model P(y | x) is often difficult, either statistically or computationally (or both), especially when x and y are high-dimensional. This is where **modeling assumptions** enter the picture.

For supervised learning problems, where we observe pairs (x,y) and ultimately seek to estimate P(y|x), there are two major categories of models: **discriminative** vs **generative**.

**Discriminative models** make assumptions about the functional form of P(y | x).
**Generative models** make assumptions about the functional forms of P(y) and P(x | y).

Consider the hospital example again:

$x_1 \cdots x_p$ symptoms, $x_j \in \{0,1\}$ e.g. "coughing"

And say that we are now tracking p binary symptoms, where p is possibly large:

$X \equiv (x_1 \cdots x_p) \in \{0,1\}^p$ takes $2^p$ possible values

e.g. $p = 40$ then $2^p \approx 1$ trillion!

We cannot simply store (or compute) p(y=1 | x) for every value of x. We need to make assumptions about the function form of p(y, x) in order to reduce that complexity.

Example of a **discriminative modeling** assumption:

Assume: $\Pr(y=1 \mid x) = \dfrac{1}{1 - \exp(-\beta^T x)}$ for some $\beta \in \mathbb{R}^p$

(This is logistic regression.)

We reduce complexity by assuming that similar x, x′ map to similar Pr(y | x), Pr(y | x′) via the function specified above.

Example of a **generative modeling** assumption: "naive Bayes"

Assume: $\Pr(x_1 \dots x_p \mid y) = \prod_{j=1}^{p} \Pr(x_j \mid y)$

In words, we are assuming that the likelihood **factorizes** across symptoms. This is equivalent to making the following **conditional independence** assumption:

$$x_j \perp\!\!\!\perp x_{j'} \mid y$$

So how does this assumption help us? Well let's compute Pr(y | x):

$$\Pr(y \mid x) = \frac{\Pr(y)\,\Pr(x \mid y)}{\Pr(x)} = \frac{\Pr(y) \prod_{j=1}^{p} \Pr(x_j \mid y)}{\Pr(x_1 \dots x_p)}$$

The first equality is due to Bayes rule, the second is due to our "naive Bayes" assumption.

What about the model evidence in the denominator? That is still high-dimensional. You might be tempted to factorize it directly, but **do not** make that mistake:

$$\Pr(x_1 \dots x_p) \neq \prod_{j=1}^{p} \Pr(x_j)$$

The naive Bayes assumption will still help us, but only via **marginalization**:

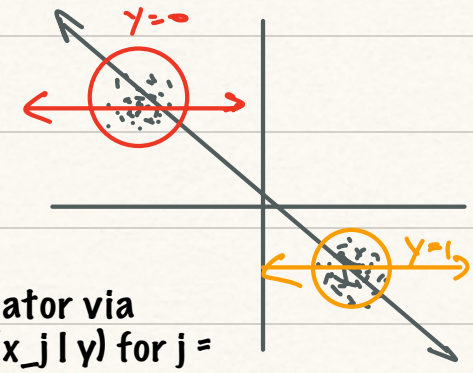
$$\Pr(x_1 \dots x_p) = \sum_y \Pr(x_1 \dots x_p \mid y) \cdot \Pr(y)$$

$$= \sum_y P(y) \prod_{j=1}^{p} \Pr(x_j \mid y)$$

The key thing to remember is that **conditional independence** is not marginal independence:

$$x_j \perp\!\!\!\perp x_{j'} \mid y \not\Rightarrow x_j \perp\!\!\!\perp x_{j'}$$

For example, this is often at the heart of **Simpson's paradox** among other famous statistical paradoxes and oddities. In the plot to the right, x1 and x2 are conditionally independent given y, but not marginally independent.




Returning to the naive Bayes model, after rewriting the denominator via marginalization, we see that we only need to estimate P(y) and P(x_j | y) for j = 1...p. This requires far fewer parameters: **2p+1, rather than 2^p.**

Is the naive Bayes assumption correct? Usually not! But the model is often useful.
"All models are wrong but some are useful." —George Box

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## Data sparsity

The parameters of the naive Bayes model are just the estimated conditional frequencies. The maximum likelihood estimate of these parameters is just the corresponding sampling frequency:

$$\text{e.g.} \quad \theta_{j1} \overset{\text{def}}{=} p(x_j = 1 \mid y = 1) \leftarrow \frac{\sum_i x_{ij} y_i}{\sum_i y_i}$$

What if our data is **sparse** such that we **never** observe x_j=1 when y=1 Then:

$$\theta_{j1} \leftarrow 0$$

This can lead to pathologies or just poor inferences:

$$p(y=1 \mid x_1 \dots x_j = 1 \dots x_p) = \frac{p(y=1) \prod_{j=1}^{p} \theta_{j1}}{p(y=1) \prod_j \theta_{j1} + p(y=0) \prod_j \theta_{j0}}$$

if $\theta_{11} = 1$, $\theta_{10} = 0$, $\theta_{21} = 0$, $\theta_{20} = 1$ then:

$$= \frac{0}{0} \quad :($$

**Laplace smoothing:** instead of setting the parameters equal to the exact sampling frequencies (which may contain exact 1s or 0s), we can first add a small constant to all the counts:

$$\theta_{j1} \leftarrow \frac{\alpha + \sum x_{ij} y_i}{2\alpha + \sum_i y_i}$$

This (very simple) fix was proposed in the 1700s by Pierre-Simon Laplace as a solution to the "sunrise problem": just because the sun has always come up in the past, doesn't mean it will 100% happen again in the future. This is the "problem of induction" in philosophy (see Bertrand Russell's book, and "Bertrand's chicken").

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# Probabilistic justification for Laplace smoothing: Beta-bernoulli inference

Although Laplace smoothing seems heuristic, it has a very clean probabilistic justification. Consider the following probabilistic model for the data:

$$X_i \overset{iid}{\sim} \text{Bernoulli}(\theta), \qquad i = 1 \dots n$$

The MLE for the Bernoulli parameter would just be the sample mean:

$$\hat{\theta}_{MLE} \overset{def}{=} \underset{\theta}{\operatorname{argmax}} \sum_{i=1}^{n} \log P(X_i \mid \theta) = \bar{X}$$

What if we introduce a prior distribution over the parameter? Assume the parameter is a beta-distributed random variable:

$$\theta \sim \text{Beta}(\alpha, \beta)$$

The beta distribution has support over the interval (0,1) and is fully described by two parameters, alpha and beta. The PDF equals:

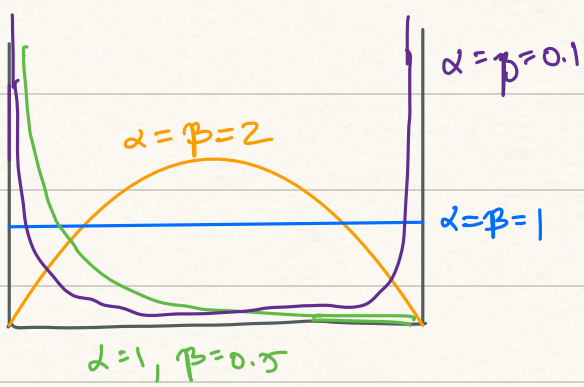
$$p(\theta \mid \alpha, \beta) = \underbrace{\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}}_{\text{normalizing constant}} \underbrace{\theta^{\alpha-1}(1-\theta)^{\beta-1}\,\mathbb{1}(\theta \in (0,1))}_{\text{kernel}}$$

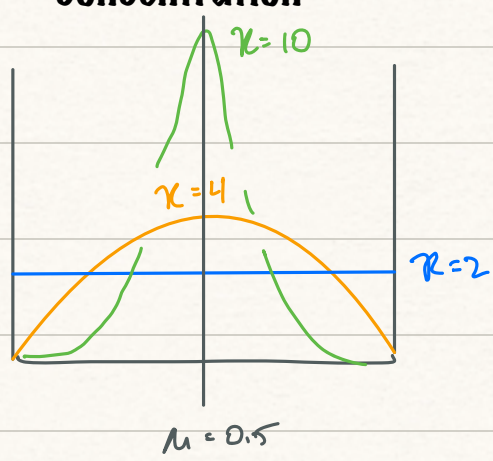
where the terms in the kernel that are proportional to theta are collectively called the **kernel**:

$$\propto_{\theta} \underbrace{\theta^{\alpha-1}(1-\theta)^{\beta-1}\,\mathbb{1}(\theta \in (0,1))}_{\text{kernel}}$$

The beta distribution can be reparameterized in terms of **mean** and **concentration**. Here are some examples of the shapes it can take:

$$\underbrace{\mu}_{\text{mean}} \overset{def}{=} \mathbb{E}[\theta \mid \alpha, \beta] \qquad \underbrace{\kappa}_{\text{concentration}} \overset{def}{=} \alpha + \beta$$



Let's calculate the posterior under the above assumptions.

$$p(\theta \mid x_1 \dots x_n) = \frac{p(\theta)\, p(x_1 \dots x_n \mid \theta)}{p(x_1 \dots x_n)}$$

We have to calculate the denominator, which is the model evidence, which equals:

$$p(x_1 \dots x_n) = \int_0^1 p(\theta)\, p(x_1 \dots x_n \mid \theta)\, d\theta$$

We could try to plug in the Beta PDF and Bernoulli PMF and evaluate this integral. However, notice that P(X1...Xn) does not involve theta. In other words, it is the **normalizing constant** of the posterior distribution. Let's first see what the kernel of the posterior is:

$$p(\theta \mid x_1 \dots x_n) \propto_\theta p(\theta)\, p(x_1 \dots x_n \mid \theta)$$

$$\propto_\theta \theta^{\alpha-1}(1-\theta)^{\beta} \prod_{i=1}^{n} x_i^{\theta}(1-x_i)^{1-\theta}$$

$$= \theta^{\alpha + \sum_{i=1}^{n} x_i - 1}\, (1-\theta)^{\beta + \sum_{i=1}^{n}(1-x_i) - 1}$$

We notice that the kernel of the posterior takes the form of the kernel of a beta distribution with the following parameters:

$$\alpha_n \overset{def}{=} \alpha + \sum_{i=1}^{n} x_i \quad , \quad \beta_n \overset{def}{=} \beta + \sum_{i=1}^{x_i} (1-x_i)$$

We are done. We know the normalizing constant for any beta distribution, which is defined in terms of the kernel. The posterior is:

$$p(y \mid x_1 \dots x_n) = \mathrm{Beta}(y \mid \alpha_n, \beta_n)$$

Posterior inference will not always be so easy. Why was it so easy this time? The answer is **conjugacy**: the Beta distribution is the conjugate prior for the Bernoulli distribution. We will explore this concept in more depth throughout the class.

The **posterior mean** is equal to:

$$\mathbb{E}[\theta \mid x_1 \cdots x_n, \alpha, \beta] = \frac{\alpha + \sum x_i}{\alpha + \beta + n}$$

Which records the Laplace smoothing scheme above when alpha=beta. Thus Laplace smoothing is justified when our prior beliefs about the parameter theta can be formalized as a symmetric beta distribution (with alpha=beta). Once such example is the uniform distribution alpha=beta=1.

Under this viewpoint, the alpha and beta parameters can be understood as "**pseudocounts**".

We can also rewrite the posterior mean using the mean-concentration parameterization:

$$= \frac{\kappa}{\kappa} \frac{\alpha}{\kappa + n} + \frac{n}{n} \frac{\sum x_i}{\kappa + n}$$

Doing so reveals the posterior mean is a weighted combination of the prior mean and the MLE:

$$= \left(\frac{\kappa}{\kappa + n}\right) \mu + \left(\frac{n}{\kappa + n}\right) \hat{\theta}_{MLE}$$

where the strength of the weight is given by the relative size of the pseudocounts (kappa = alpha + beta) to the number of data points n. The concentration parameter kappa is thus the **strength of the prior belief**.

## Subjectivist interpretation of probability

**What is p(theta)? It is not a frequency, since there is only one theta. The subjectivist approach says instead that it represents a degree of belief.** This is a different use/interpretation of probability than is common in standard frequentist statistics; this difference is at the heart of the famous "Bayesian versus frequentist" debate in prior decades.

Think about the following commonplace uses of probability:

- "What's the probability the Bulls win on Thursday?"
- "What would you say is the probability that the Riemann conjecture is true?"

These questions involve the use of "probability" which cannot be understood (well) as a frequency: these questions ask about degrees of belief.

# The Dutch book argument

The use of probability to quantify degrees of belief is not arbitrary. In fact, we can **prove** (in some sense) that **the only coherent way to represent beliefs is via probability.** This is what is known as the "Dutch book argument".

Here's the setup. You and a Gambler are making bets about the occurrence of events which are defined over a sample space Sigma.

Let F be a collection of events defined over Sigma that is:

- closed under unions: $A, B \in \mathcal{F}$ then $A \cup B \in \mathcal{F}$
- closed under complement: $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$

You are the bookie. Your job is to specify a **pricing function:**

$$\mu(A) \in \mathbb{R} \quad \forall A \in \mathcal{F}$$

You are free to specify this function any way you want. However, once you have done so, the Gambler then gets to propose a wager consisting of a set of bets of the following form:

"Bet $a on A" means that the Gambler wagers (i.e. pays you) $a mu(A), and if the event A actually does in fact occur, then you pay the Gambler $a. Your net winning on that bet is thus:

$$a(\mu(A) - \mathbb{1}(A))$$

The Gambler can also make **conditional bets** of the form: "Bet $b on the event B conditional on A" which means the Gambler pays you $b mu(A | B) if A occurs and then if B also occurs you pay the Gambler $b. Your net winnings in this case is:

$$\mathbb{1}(A)\, b\, (\mu(B|A) - \mathbb{1}(B))$$

Where no money ever changes hands if A does not occur. Also note that you are being asked to additionally specify the pricing function for conditioning events defined for any pair A,B in F.

Again, there are no constraints on how you set this pricing function. However, as it turns out, unless you are very careful with how you set the pricing function, the Gambler will be **guaranteed to make money.**

Let's see a simple example of that. The set F contains all subsets and their complements the of sample space including the entire sample space Omega the null set (its complement)

$$\Omega, \phi \in \mathcal{F}$$

How should you price these? Well Omega is the union of all events, **which always occurs**. Your net winning on the wager "Bet 1 on Sigma" is

$$\mu(\Omega) - \mathbb{1}(\Omega) = \mu(\Omega) - 1$$

So you should definitely set mu(Sigma) >= 1, otherwise the Gambler can **always** make money by betting 1 on Sigma.

Now say you set mu(Sigma) = 2. The Gambler can propose negative bets e.g., "Bet -1 on Sigma" which would then guarantee that you lose 1.

As it turns out, unless you set mu(Sigma)=1, the Gambler will always have a bet that is guaranteed to make money against you. By essentially the same argument, you must also set mu(null set)=0.

In general, a sequence of bets that are **guaranteed to win money** is called a Dutch book. (The origin of that term is disputed and not known.) A pricing function against which a Dutch book can be constructed is then called **incoherent**. Your job as a bookie is to create a coherent pricing function.

What else can be said about coherent pricing functions? One can prove that a pricing function must obey the following conditions to be coherent:

1. $0 \leq \mu(A) \leq 1 \quad \forall A \in \mathcal{F}$

2. $\mu(A \cup B) = \mu(A) + \mu(B) \quad \text{if } A \cap B = \emptyset$

   ↳ This implies $\mu(A) = 1 - \mu(A^c)$

3. $\mu(A \mid B)\mu(B) = \mu(A \cap B)$

As it turns out, these conditions on the pricing function being coherent exactly define a probability measure! This is one of the famous theorems of de Finetti (from the 1930s):

Theorem (de Finetti): Suppose mu is a pricing function defined over (Sigma, F). mu is coherent IFF mu is a finitely additive probability measure over (Sigma, F).

De Finetti was famous for formalizing the "subjectivist" interpretation of probability. This theorem informally shows that any system of beliefs which cannot be quantified by a probability measure is "incoherent" in the sense that it is possible to be **guaranteed** to win money against it. (Note: A pricing function can be both coherent, and **very unlikely** to win money if the probabilities to not match the true empirical probabilities.)